

Research Paper

Study on heating capacity and heat loss of capillary radiant floor heating systems

Pei Ding^a, Yanru Li^b, Enshen Long^{a,b,c,*}, Yin Zhang^b, Qinjian Liu^b^a Institute for Disaster Management and Reconstruction, Sichuan University-Hong Kong PolyU, Chengdu, China^b College of Architecture and Environment, Sichuan University, Chengdu, China^c Key Laboratory of Deep Earth Science and Engineering (Sichuan University), Ministry of Education, Chengdu, China

HIGHLIGHTS

- A new simplified calculation method for radiant floor was proposed.
- Calculated results agree well with measured and numerical simulated data.
- Floor equivalent thickness affect heating capacity varied with pipe spacing.
- Pipe wall with twice heat conductivity than filled layer can be ignored.

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ABSTRACT

Radiant floor heating systems with the distinct advantages of low-temperature water supply, energy saving, uniform room temperature and well thermal comfort, are widely used in residential and office buildings in China. And those radiant floors embedded with capillary pipes have been increasing in popularity because of better thermal performance. In this paper, a new simplified calculation for heating capacity, heat loss and surface temperature of radiant heating floor with high-performance insulation layer was provided and validated by measured as well as numerical simulated data. By the simplified calculation method, the influence of each layer material on the heat transfer of capillary radiant heating floor have been analyzed and results show that, (1) the heating capacity will decreased with the increasement of pipe spacing and the descending slop is decided by the ratio of pipe spacing to the equivalent distance from pipes to the room environment; (2) thermal resistance of pipes could be ignored when it is twice more than that of filling material, or it will have negative influence on heating capacity with the increase of pipe spacing.

1. Introduction

Building energy conservation contributes an important part of energy saving potential worldwide [1]. In 2015, the total building energy consumption of China was 857 million tons of standard coal, accounting for 19.93% of total energy consumption. And the energy consumption of central heating in the northern China was 193 million tons of standard coal, which showed a rising trend year by year [2]. Therefore, the research and optimization of heating system play an important role in energy conservation. Compared to traditional air distribution systems or high and medium temperature radiator systems, low-temperature hot water radiant floor heating systems are more energy efficient and comfortable [3]. As for radiant floor heating systems, the heat exchange between radiant floor and indoor thermal

environment is mainly by radiation, so that the room temperature required to achieve thermal comfort could be lower [4]. It also could reduce local discomfort in the foot area, because of less temperature fluctuations and temperature gradients in vertical direction [5].

Capillary radiant floor systems have the distinction of smaller pipe diameter and spacing [6]. In virtue of above characteristics, they are more flexibility to be installed in floors [7], walls [8] or ceilings [9]. Furthermore, its thinner structure makes room temperature response be more rapid [10,11]. Xie et al. [12] used computational fluid dynamics method to study cooling capacity of capillary ceiling radiant cooling panel and found it had negative linear correlation with temperature of chilled inlet water, covering thickness and tube spacing. Mikeska et al. [13] numerically investigated the thermal conductivity sensitivity of concrete to heat flux, and results showed that it depended on the

* Corresponding author.

E-mail address: longes2@163.com (E. Long).<https://doi.org/10.1016/j.applthermaleng.2019.114618>

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Nomenclature

A	area (m ²)
$AUST$	average unheated surface temperature (°C)
Bi	Biot number (dimensionless)
d	distance (m)
H	thickness (m)
h	heat transfer coefficient (W/m ² /°C)
L	pipe spacing (m)
l	length of pipe (m)
Pr	Prandtl number (dimensionless)
q	heat flux (W/m ²)
R	thermal resistance (m ² ·°C/W)
Re	Reynolds number (dimensionless)
T	temperature (°C)
U	overall heat transfer coefficient (W/m ² /°C)
δ	internal diameter of pipe (m)
θ	excess temperature (°C)
λ	thermal conductivity (W/m/°C)

σ	external diameter of pipe (m)
φ	reduction ratio of heat flux (dimensionless)

Subscript

0	center line
1	upper part
2	lower part
c	convection
f	filling layer
fs	floor slab
is	insulation layer
o	operative temperature
m	mean value
p	pipe
r	radiation
s	surface layer
w	water

configuration of the capillary micro tubes. Zhao et al. [14,15] set up a solar phase change thermal storage heating system using a radiant-capillary-terminal, and air source heat pump as assistant heating appliance. It was concluded that the heating system energy saving rate could be significantly improved by lowering the water supply temperature.

Over the past decades, many scholars have studied the heat transfer models of radiant floor [16]. Since the thermal network models can provide simplified calculation procedures and relatively high calculating precision, it has received many attentions. Koschenz and Lehmann [17,18] developed a one-dimensional model to calculate the heating capacity and heat loss of concrete slab heating/cooling system, based on analytical solution [19]. Tao [20] studied the low temperature radiant system module integrated in EnergyPlus, and modified the module with calculated data. Assuming that heat transfers upward and downward separately from the horizontal center line of pipe layer, Wu [21] proposed a new simplified calculation by using shape factor method. The calculated surface temperature and heat flux both agreed well with experimental and simulated data. Heat loss of the building envelope with radiant floor might be higher than that without integrating radiant heating system [13], and it also may cause the failure to reach the comfortable temperature or the lower room to be overheated [22]. Optimizing the thermal insulation of buildings is an effective way to reduce heat loss [23]. Zhang et al. [24,25] established a simplified calculation model of radiant floor ignoring the heat loss downward the floor.

In engineering application, simplified thermal model can be used for calculating heating capacity and surface temperature of radiant heating floor. At present, utilizing insulation material to reduce heat loss is a common design in buildings. Thus, this paper provides a new simplified calculation based on previous studies to quick estimate heat transfer of radiant floor. By the simplified calculation, factors affecting capillary radiant heating floor are analyzed. And the results could provide a reference for thermal performance estimation and design parameters optimization of capillary radiant floor systems.

2. Simplified calculation of heat transfer for radiant floor

2.1. Governing equations and analytical solution

Assuming that temperature along the water pipe is consistent, the 2D steady-state heat transfer of radiant floor shown in Fig. 1 is governed by Eq. (1). Define excess temperature as $\theta = T - T_{o1}$, where T_{o1} is the indoor operative temperature.

$$\frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2} = 0 \quad (1)$$

The boundary conditions are:

Around the water pipes, $(x - nL)^2 + y^2 = r^2$, where n the numbering of pipes

$$\theta = \theta_p \quad (2)$$

On the upper floor surface, $y = d_1$

$$\lambda \cdot \frac{\partial \theta}{\partial y} + h_1 \cdot \theta = 0 \quad (3)$$

On the lower floor surface, $y = -d_2$

$$\lambda \cdot \frac{\partial \theta}{\partial y} + h_2 \cdot \theta = 0 \quad (4)$$

When there is well-insulated below the pipe layer, the distance d_2 is regarded as approaching to ∞ . Then, temperature distribution inside radiant floor could be expressed as Eq. (5) [19].

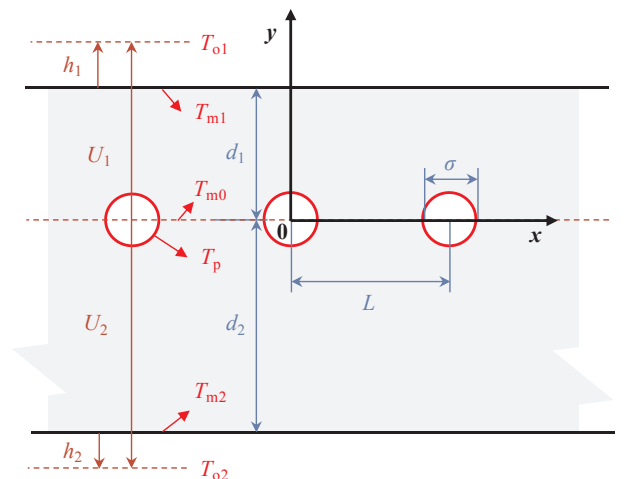


Fig. 1. 2D schematic of radiant floor.

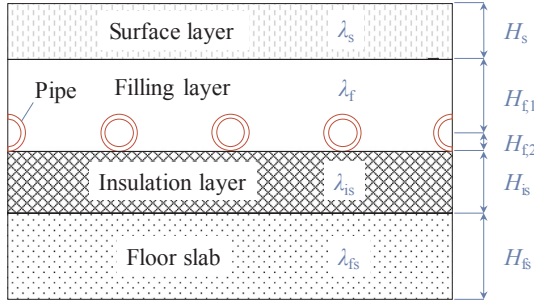


Fig. 2. Schematic of radiant floor with multilayer structure.

$$\theta(x, y) = -\Gamma \cdot \theta_p \cdot \left\{ \frac{2\pi}{L} \cdot \left(y - \frac{\lambda}{U_1} \right) - \sum_{s=1}^{\infty} \frac{1}{s} \cdot [e^{-(2\pi s/L)y} + g(s) \cdot e^{(2\pi s/L)y}] \cdot \cos\left(\frac{2\pi s}{L} \cdot x\right) \right\} \quad (5)$$

$$\Gamma = \left[\ln\left(\frac{L}{\pi\sigma}\right) + \frac{2\pi\lambda}{L \cdot U_1} + \sum_{s=1}^{\infty} \frac{g(s)}{s} \right]^{-1} \quad (6)$$

$$g(s) = -\frac{Bi - 2\pi s}{Bi + 2\pi s} \cdot e^{-(4\pi s/L) \cdot d_1} \quad (7)$$

$$Bi = h_1 \cdot L / \lambda \quad (8)$$

Mean temperature of pipe center line ($y = 0$) and floor surface ($y = d_1$) could be obtained by Eqs. (9) and (10) respectively. Then, heat flux of upper floor surface could be calculated by Eq. (11).

$$\theta_{m0} = \int_0^L \theta(x, 0) dx = \Gamma \cdot \frac{2\pi\lambda}{L} \cdot \frac{\theta_p}{U_1} \quad (9)$$

$$\theta_{m1} = \int_0^L \theta(x, d_1) dx = \Gamma \cdot \frac{2\pi\lambda}{L} \cdot \frac{\theta_p}{h_1} \quad (10)$$

$$q_1 = \theta_{m0} \cdot U_1 = \frac{2\pi\lambda \cdot \Gamma}{L} \cdot \theta_p \quad (11)$$

2.2. Thermal network model

For radiant floor with multilayer structure in Fig. 2, thermal resistance of each layer is calculated by Eq. (12), and the heat transfer rate is equal to the temperature difference divided by the thermal resistance between pipes and room environment (Eq. (13)).

$$R_i = H_i / \lambda_i \quad (12)$$

where i represents surface layer ($i = s$), filling layer ($i = f$), insulation layer ($i = is$) and floorslab layer ($i = fs$).

$$q_1 = \theta_p / R_{p1} \quad (13)$$

As shown in Fig. 3, it is assumed that there is a virtual layer between pipes and the filling layer [20]. By Eqs. (6), (11), (12) and (13), thermal resistance of the virtual layer could be calculated by Eq. (14).

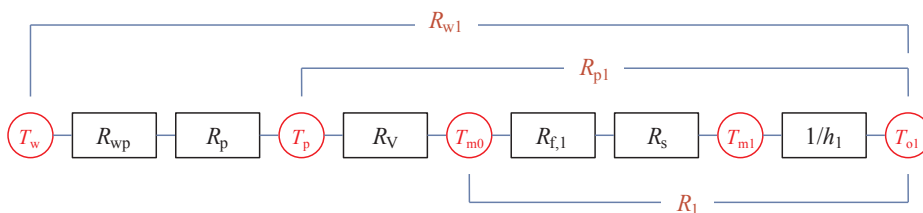


Fig. 3. Structure schematic of thermal resistances in radiant floor.

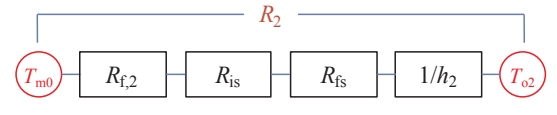


Fig. 4. Thermal resistance of radiant floor below pipe layer.

Table 1

Thermal parameters of each layer in the test piece.

Structure	Material	Thermal conductivity [W/(m °C)]	Thickness (mm)
Surface layer	—	—	—
Filling layer	Cement mortar	0.93	20
Capillary pipes	PE-RT	0.4	3.4 × 0.55
Insulation layer	Polystyrene foam	0.04	20

Notes: thickness for pipes, external diameter × pipe wall thickness.

$$R_v = \frac{L}{2\pi\lambda_f} \cdot \left[\ln \frac{L}{\pi\sigma} + \sum_{s=1}^{\infty} \frac{g(s)}{s} \right] \quad (14)$$

Operative temperature has been defined to evaluate indoor thermal environment, which is calculated by Eq. (15). For the floor upper surface, the convective heat transfer coefficient can be obtained by Eq. (17) [26]. the radiant heat transfer coefficient could be considered as a constant of 5.5 W/(m² °C) [27]. The average unheated surface temperature ($AUST$) is the arithmetic average temperature of indoor non-heated wall surfaces. Usually, MRT (mean radiant temperature) instead of $AUST$ is used to calculate operative temperature [28]. While in an acceptable thermal environment, they could be thought as approximately equal because floor surface temperature is required to less than 29 °C [29].

$$T_{o1} = (h_c \cdot T_{air} + h_r \cdot AUST) / h_1 \quad (15)$$

$$h_1 = h_c + h_r \quad (16)$$

$$h_c = 2.13(T_{m1} - T_{air})^{0.31} \quad (17)$$

$$AUST = \frac{\sum_{j=1}^n A_j T_j}{\sum_{j=1}^n A_j} \quad (18)$$

When thermal resistances between thermal fluid and pipes are considered, the heat flux will be expressed as Eq. (19), as shown in Fig. 3. The thermal resistance between the thermal fluid and the pipe wall can be obtained by Eq. (20). And the thermal resistance of pipe wall is calculated by Eq. (22).

$$q_1 = \theta_w / R_{w1} \quad (19)$$

$$R_{wp} = \frac{L}{\pi\sigma} \cdot \frac{1}{h_{wp}} \quad (20)$$

$$h_{wp} = 1.86 \left(\frac{Re \cdot Pr}{l/\delta} \right)^{1/3} \frac{l}{\delta} \quad (21)$$

$$R_p = \frac{L}{\pi\sigma} \cdot \frac{H_p}{\lambda_p} \quad (22)$$

By Eq. (10), the mean temperature of pipe center line can be

Table 2
Operating parameters of tests.

Cases	Supply water temperature(°C)	Indoor design temperature (°C)	Measured average temperature of supply and return water (°C)	Flow rates (m/s)	Pipe spacing (m)
1	34	20	31.1	0.15	0.02
2	34	20	32.3	0.25	0.02
3	34	20	32.8	0.35	0.02
4	34	20	33.1	0.15	0.01

Table 3a
The upper surface heat flux of capillary radiant heating floor (Unit: W/m²).

Case	Measured	Calculated	Error	Simulated	Error
1	79.4	82.5	3.1	80.1	0.7
2	94.9	90.3	-4.6	88.9	-6.0
3	97.3	94.7	-2.6	91.9	-5.4
4	101.2	106.1	4.9	105.1	3.9

Table 3b
The upper surface temperature of capillary radiant heating floor (Unit: °C).

Case	Measured	Calculated	Error	Simulated	Error
1	29.2	29.5	0.3	29.2	0
2	30.9	30.5	-0.4	30.3	-0.6
3	31.2	30.9	-0.3	30.6	-0.6
4	30.5	31.0	0.5	30.9	0.4

Table 3c
The lower surface heat flux of capillary radiant heating floor (Unit: W/m²).

Case	Measured	Calculated	Error	Simulated	Error
1	20.3	17.9	-2.4	17.4	-2.9
2	20.1	19.4	-0.7	18.9	-1.2
3	23.7	20.2	-3.5	19.5	-4.2
4	20.9	22.2	1.3	21.9	1.0

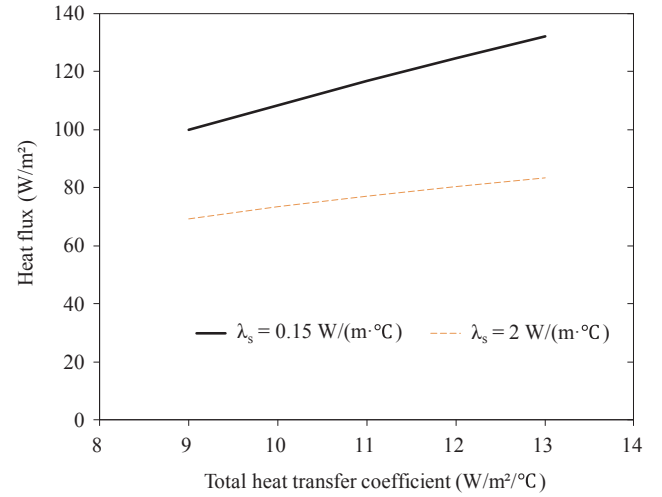


Fig. 6. Effect of surface heat transfer coefficient on heat flux.

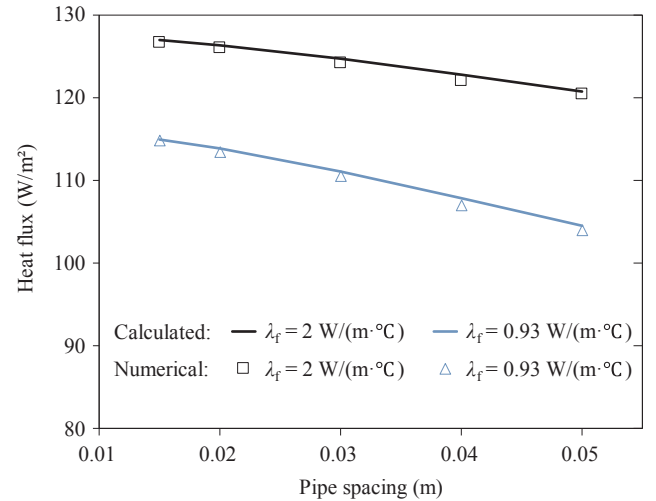


Fig. 7. Effect of filling material and pipe spacing on heat flux.

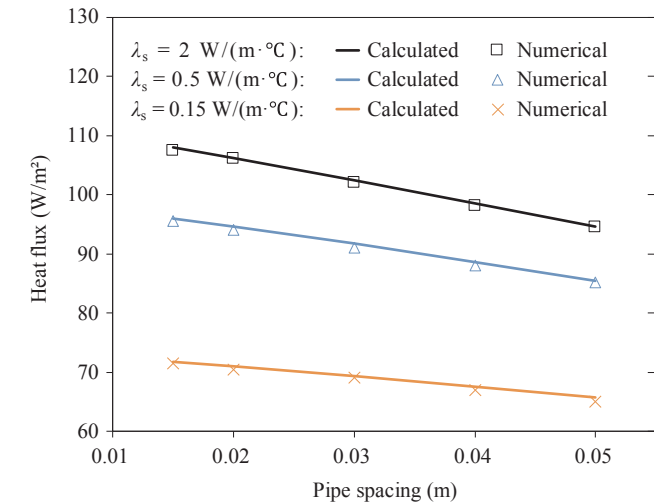


Fig. 5. Effect of surface material and pipe spacing on heat flux.

obtained. Then, the heat loss through the lower floor surface could be calculated by Eq. (23), as shown in Fig. 4.

$$q_2 = q_1 \cdot \frac{R_1}{R_2} \cdot \frac{\theta_{m0} - \theta_{o2}}{\theta_{m0}} \quad (23)$$

3. Validation with experimental and simulated data

3.1. Experimental cases

To verify the established simplified calculation model, measured data from references [30] are cited here. The experiments were carried out on a low temperature hot water standard test bench, in a 4 m × 4 m × 2.8 m hexahedral closed chamber. Outside of the chamber, an air conditioning system was used to keep the indoor air temperature as a constant. And a water system consisting of a water pump, water tanks, a heater, and a flow rate test device provides hot water with different constant temperature for the capillary end. Then, 8 thermocouples and heat flux sensors were arranged on the upper and lower surface of the test piece, to measure the temperature and heat flux data till a steady state. The thermal parameters of each layer are

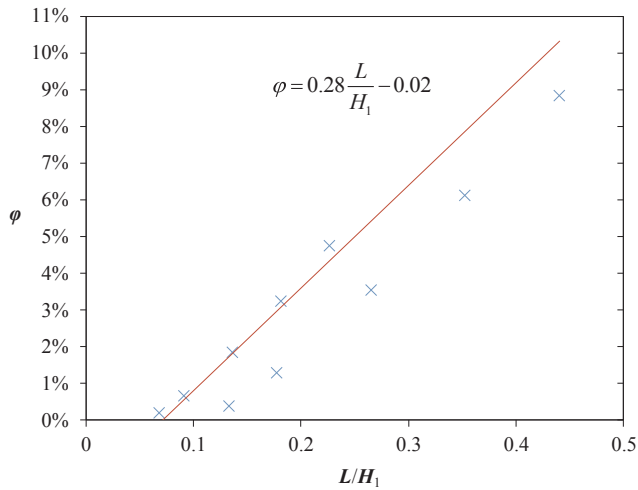


Fig. 8. Effect of dimensionless parameter L/H_1 on heating capacity reduction.

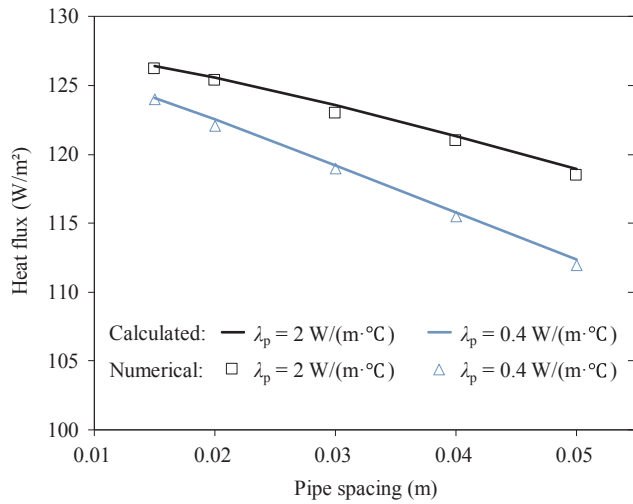


Fig. 9. Effect of pipe material and pipe spacing on heat flux.

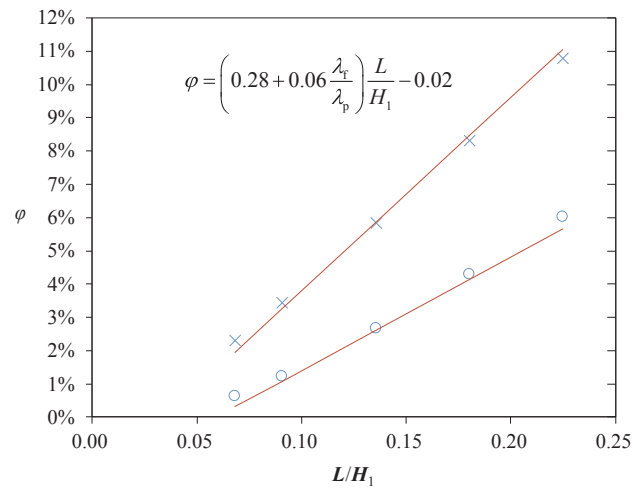


Fig. 10. Effect of dimensionless parameter L/H_1 and λ_t/λ_p on heating capacity reduction.

shown in Table 1. The heat flux and temperature of the test piece surfaces were measured under 4 operating conditions, as shown in Table 2.

3.2. Numerical simulation model

The finite difference method (FDM) has been widely used to simulate heat transfer of radiant floor. In this paper, 2D steady-state numerical simulation models as Fig. 2 have been built in ANSYS Fluent software to validate the simplified calculation model. And the following assumptions are made:

- (1) Materials of each layer are homogeneous and contact resistances are negligible.
- (2) The left and right side boundaries are regarded as adiabatic, due to the symmetric distribution of pipes.
- (3) Ignoring the thermal resistance between thermal fluid and pipes, temperature distributes uniformly along the pipe interior wall.

3.3. Results comparison

Tables 3 shows the comparison of measured data with calculated and numerical simulated data under four different operating conditions. The relative errors between measured and calculated upper surface heat flux are within 5%. The errors between measured and calculated are within $\pm 0.5^\circ\text{C}$ for upper surface temperature, and within $\pm 3.5\text{ W/m}^2$ for lower surface heat flux. The errors between measured and calculated data are mainly from the assumptions of simplified calculation model as well as the measuring errors. The simulated data are slightly lower than calculated data, and also agree well with measured data. Thus, the simplified calculation method established in this paper was reliable, and that is available to use it to estimate the thermal performance of radiant floor in engineering application.

4. Discussions

4.1. Effect of surface material on heating capacity

Surface layer is an important factor affecting upward heat transfer of the heating floor, which is determined by the thickness and thermal conductivity of its component material. As shown in Fig. 5, the influence of surface material thermal conductivities from 0.15 to $2\text{ W/(m}\cdot^\circ\text{C)}$, as well as pipe spacing with a range of 0.015 m to 0.05 m on heating capacity was analyzed. It could be seen that heat flux increases with the increment of thermal conductivity. The heat flux decreases approximate linearly with the increment of pipe spacing. And the descending rate is decided by both the thermal conductivity and pipe spacing. The higher thermal conductivity of surface material, the greater influence of pipe spacing on heat flux.

4.2. Effect of surface heat transfer coefficient on heating capacity

The total heat transfer coefficient between floor upper surface and indoor environment is the dominant limitation of heating capacity [24]. As for radiant heating floor, it is sometimes regarded as a constant of $10.8\text{ W/(m}^2\cdot^\circ\text{C)}$ [31]. Fig. 6 shows the effect of total heat transfer coefficient on heat flux of capillary radiant heating floor. It could be seen that, the slight change of heat transfer coefficient is able to lead to non-negligible influence on heat flux, especially when surface material thermal conductivity is relatively high. Thus, that is of great importance for refined calculation of heating capacity choosing precise surface heat transfer coefficient [26]. On the other hand, enhancing the total heat transfer coefficient would obviously improve the heating capacity or lower the water supply temperature required [32,33].

4.3. Effect of filling material on heating capacity

Fig. 7 shows the influence of filling material on heating capacity of capillary radiant heating floor when thermal resistance of pipes is

ignored. Being the same as surface material, filling material with higher thermal conductivity can increase the heating capacity through upper floor surface. However, as the increase of thermal conductivity, the influence of pipe spacing on heat flux become smaller, which is different from surface material.

To quantify the influence of filling material and pipe spacing on heat transfer of radiant floor, define φ as the reduction ratio of heat flux. As shown in Fig. 8, dimensionless parameter φ and L/H_1 present linearly relationship, where H_1 is the equivalent distance from pipe center line to room environment, calculated by Eq. (24).

$$H_1 = \lambda_f \cdot R_1 \quad (24)$$

$$q_1 = (1 - \varphi) \cdot \theta_w / R_1 \quad (25)$$

4.4. Effect of pipe material on heating capacity

Fig. 9 shows the influence of pipe material on heat flux. When the thermal conductivity of pipe material changes from 0.4 W/(m °C) to 2 W/(m °C), heat flux has increased slightly. But with the increase of pipe spacing, the heat flux difference increases. That is, pipe material will affect the slope of heating capacity reduction with pipe spacing increase.

As shown in Fig. 10, the dimensionless parameter λ_f/λ_p would affect the reduction ratio of heating capacity. The thermal resistance of pipe material could be ignored while its thermal conductivity is high enough [34]. For capillary radiant floor heating system, if the thermal conductivity of pipe material is two times than that of filling material, then the thermal resistance of pipes could be considered as negligible, because the filling material would play a decisive role on heating capacity reduction.

5. Conclusions

In this paper, a new simplified method for estimating the heating capacity, surface temperature and heat loss of radiant heating floor was established. And the calculation method was verified with experimental and numerical simulated data for capillary radiant floor heating systems. Then, those factors affecting its heating capacity have been analyzed, and important conclusions are as following.

- (1) As pipe spacing increases, heating capacity and heat loss of the systems decrease linearly. The descending slope is related to the ratio of pipe spacing to the equivalent distance from pipes to room environment.
- (2) The thermal conductivity of pipe material also has influence on the descending slope. When it is about twice more than filling material thermal conductivity, the influence could be negligible.
- (3) Enhancing total heat transfer coefficient between floor surface and room environment can improve the heating capacity, and the effect is affected by the thermal resistance of surface material.

Declaration of Competing Interest

We declare that we have no conflict of interest.

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